PRG1781 Assessment: Assignment 2

Due date: 27/03/2026

Total: 45 Marks

Introduction

Algorithms are fundamental to how modern computer systems process and analyze data. Many software applications rely on efficient algorithms to organize, evaluate, and manage information.

Question 1 Subtotal: [40 Marks]

1.1 Sorting Algorithm Analysis (10 Marks)

Explain using a step-by-step explanation and a decision tree diagram how a Merge Sort algorithm will sort the following list: **{45, 20, 60, 15, 35, 50, 10}**

1.2 Algorithm Design Using Mathematical Formula (10 Marks)

An online store calculates delivery cost using: **TotalDeliveryCost = BaseFee + (Distance × CostPerKm)** Break down the formula into logical steps and develop pseudocode with comments.

1.3 Real-World Algorithm Development Scenario (20 Marks)

Develop an algorithm for a university student attendance monitoring system that stores attendance linked to student numbers. Display 'Present' if attending and 'Absent' if not. Include pseudocode and a flowchart and ensure multiple records are processed.

Presentation: [5 Marks]

Marks are awarded for structure, clarity, readability, correct diagrams, and professional formatting.

Criteria	Marks
1.1 Merge Sort Explanation	10
1.2 Formula Pseudocode	10
1.3 Attendance Algorithm	20
Presentation	5
Total	**45**